Project Meeting Notes

ENGSCI 700 — Wind-Assisted Cargo Vessel Routing

Week 4 Meeting

Focus: Problem Scoping and Data Strategy

- Discussed simulating routing using historic weather data, combining expected weather with historically optimal routes.
- Proposed solving the routing problem backwards from the destination (known arrival time), though this was noted as a less favoured direction.
- Identified opportunity to examine timestamp data comparing wind-assisted versus non-wind-assisted voyages to find empirical evidence of routing benefits.

Literature Review Goals:

- Demonstrate breadth of literature engagement.
- Identify the key influences on routing decisions that will inform modelling choices.
- Define project scope clearly.

Action Items:

- Examine existing codebase and attempt to run it.
- Understand how the existing code handles weather data.
- Begin background comparison of wind-assisted and conventional routes.

Week 5 Meeting

Focus: Understanding Existing Tools and Proposing Methodology

- Investigated how OpenCPN and PredictWind perform routing (both appear to use grid-based minimum cost path algorithms, likely with dynamic programming).
- Discussed downloading large volumes of Atlantic Ocean weather data and running the existing Dijkstra's algorithm.
- Identified the need to recompute routes mid-voyage as forecasts change.

Proposed Methodology:

- Collect daily PredictWind forecasts over a target time period (up to 15 days ahead, 3–24 hourly resolution), treating each model as an ensemble member.

- Solve the shortest path under each forecast; resimulate daily as new forecasts arrive.
- Evaluate the policy by replaying the voyage day by day under successive forecasts.

Technical Notes:

- PredictWind offshore app allows area-specific GRIB downloads (standard subscription required).
- PredictWind's own router uses ensemble forecasting; exact algorithm unclear but appears grid-based with dynamic programming.
- Explored using A* to improve computational efficiency.
- Identified potential to automate GRIB file downloads via PyAutoGUI.

Meeting — Data Preprocessing and Policy Design

Focus: Efficiency and Initial Policy Concepts

- Discussed preprocessing GRIB data into in-memory structures to avoid repeated hard-disk reads.
- Proposed caching great-circle distances and longitude/latitude differences rather than recomputing haversine functions each call.
- Outlined approach: sail on the current heading for a forecast period (24 hours), compute average speed required to reach that point under each ensemble member, then run Dijkstra's from that point under each forecast for the remainder of the voyage. Resimulate daily.

Technical Actions:

- Pre-process and compress GRIB files into simpler in-memory readings.
- Keep files open for the duration of a solve rather than repeatedly opening and closing.
- Investigate memory-mapped files (Python `mmap` module) for efficiency.
- Profile the code to understand timing bottlenecks.

Meeting — Grid Refinement and Interpolation

Focus: Fixing Grid Distortions

- Identified an issue where extended-neighbourhood arcs assume constant wind from the current node, rather than picking up intermediate grid-point wind conditions — effectively exploiting the discretisation architecture.
- Proposed interpolating polars and weather data between grid points in space and time to correct this.
- Discussed using 3-hour intervals as the forecast update frequency; changing weather data at the 3-hour mark within a 12-hour cycle.

- Agreed to refine resolution until a convergence tail-off is observed.

Action Items:

- Code up an interpolation scheme for use in both the Dijkstra's algorithm and the new policy.
- Redefine boundaries around peninsulas to match the coarsest grid, ensuring consistent land masking across resolutions.
- Investigate memory-mapped file access for GRIB sharing across parallel policy solves.

Meeting — Synthetic Network and Policy Intuition

Focus: Building the Synthetic MPC Framework

- Proposed constructing a simplified layered network (directed acyclic graph) with random arc costs as a test bed for the new policy.
- Network structure: start node → columns of intermediate nodes → end, with forecast updates when each column of nodes is reached.
- Discussed using northerly vs. southerly wind scenarios with correlated arc-cost evolution (scenarios correlated within a forecast model, updated when a new column is reached).
- Key question posed: does the adaptive policy degenerate to a greedy strategy under certain conditions?

Comparisons to Run:

- Adaptive policy vs. a single fixed forecast (deterministic plan) under common random numbers.
- Vary forecast noise and persistence to understand when adaptive routing outperforms a fixed plan.

Meeting — Formalising the Stochastic Framework

Focus: Policy Evaluation Design

- Formalised the fixed-forecast vs. adaptive comparison: run both under shared Monte Carlo realisations (common random numbers) to isolate the benefit of adaptivity.
- Defined test parameters: evolving forecast noise — uniform, slow-then-sudden switch, or fast-changing.
- Agreed to write up methodology fragments iteratively, sharing with supervisors as work progresses rather than leaving all writing to the end.

Meeting — Isochrone Routing and Adaptive Step Size

Focus: Expanding the Routing Approach

- Proposed investigating an isochrone-based weather routing approach (as used in Sail Router).
- Isochrones define all reachable points within a fixed time window (e.g., 2–2.1 days), providing a richer set of headings than a fixed grid.
- Discussed adaptive step sizing: finer steps near land or in rapidly changing weather; coarser steps in open ocean under stable conditions.
- Proposed colour-coding nodes by proximity to land and weather variability to determine step size dynamically.
- Discussed convergence check: test at least three discretisation levels to confirm that routes are insensitive to grid resolution.

Further Actions:

- Explore motor/engine energy usage and its discretisation.
- Simulate under actual (realised) weather to validate policy routes.
- Confirm that the adaptive policy always outperforms a purely random baseline.
- Compute the best-oracle (hindsight-optimal) route by working backwards from the realised weather, to serve as a lower bound.

Meeting — Grid Consistency Across Resolutions

Focus: Multi-Resolution Validation

- Agreed that for the three coarsest discretisations, land masking should be consistent: any ocean point inside the coarsest grid's land cell is treated as land across all resolutions.
- Plan to simulate routes at different arc radii on the same example and compare travel durations.
- Identified need to count the number of nodes processed and runtime as a measure of computational effort.
- Discussed profiling edge relaxation and arc-cost computation as the main bottleneck for longer arcs.

Meeting — GRIB Validation and Presentation Preparation

Focus: Validation Milestones and Presentation Strategy

Completed milestones:

- Validated results using GRIB data ✓
- Implemented backward DP (flipped graph) ✓
- Created interactive Plotly visualisations ✓
- Completed additional arc-cost measure testing ✓

Presentation Approach:

- Present the core problem and highlight key results.
- Avoid deep dives into the stochastic policy framework (too complex for the allotted time).
- Pending: test under actual (ERA5) weather; add active-measure panel to the interactive plot.

Meeting — Rolling Horizon and MPC Framework

Focus: Framing the Stochastic Problem

- Clarified that the core problem is a rolling-horizon (receding-horizon) stochastic optimisation problem: we never observe the true weather process directly, only forecast realisations.
- All policies are therefore heuristics, since the true stochastic process governing weather is unknown.
- Formally adopted the Model Predictive Control (MPC) framing rather than a one-shot dynamic programme.
- Proposed incorporating climatology as an alternative or supplementary forecast source, especially beyond the 10-day PredictWind forecast horizon.

Meeting — Forecast Weighting and Discounting

Focus: Policy Refinements

- Proposed weighting forecasts by their cost-to-go values (up-weighting the forecast that looks most costly or most promising looking ahead).
- Discussed discounting arc costs by lead time (e.g., weight = 1 for the immediate arc, 0.9 for the next stage) to emphasise near-term certainty.
- Proposed CVaR (Conditional Value-at-Risk) analysis for risk-averse policy variants — deferred to future work.
- Discussed using a convex combination of discounted forecasts with climatology — deferred to future work.

Immediate Actions:

- Implement actual start and end nodes in the synthetic network diagram; set inter-node spacing to approximately one day's sailing distance for a cargo vessel.

- Test the adaptive and greedy policies on a deterministic (static) wind field to diagnose behavioural differences.
- Code up the arc-cost discounting scheme.

Meeting — Greedy Diagnosis and Policy Separation

Focus: Understanding Greedy Failure Modes

- Confirmed that completely different routes are produced by different deterministic forecasts.
- Introduced two-scenario switching experiments to study the transition from deterministic to stochastic conditions.
- Proposed reducing engine power to create greater policy separation in results — at lower engine power, wind has more influence on routing decisions.

Report Guidance:

- Report results as percentage deviation from the hindsight optimal (hindsight gap Δ^p).
- For the final report, select 20 random days; use climatology-aligned days for targeted comparison figures.
- Review last year's reports for engine speed experiments and compare findings.

Report Writing Notes

- Reference style: square brackets (numbered) throughout.
- In the literature review, use a figure for the polar diagram section rather than prose description; remove padding.
- Simplify the sail types section: keep rotor sails; merge remaining sail types into one short paragraph.
- Remove the A* policy discussion from the report.
- Remove all em-dashes throughout the document.
- In Section 2.6.2, describe the approach as 'my approach' rather than 'the problem'.
- Explain superscript notation at the top of page 18.
- Use figures to shorten explanations where possible.
- Introduce the terms 'receding horizon' and 'MPC framework' explicitly at first use.
- Change the calligraphic F to capital Omega (Ω) for the forecast ensemble notation.
- Write as if the reader has not read the section headings — guide the reader through the narrative.
- Use 'simulation' (not 'run' or 'experiment') consistently when describing the policy loop; a policy is a decision rule, not the outcome of a simulation.
- Order the report chronologically — in the order experiments were conceived and run — rather than by a traditional methodology-then-results structure.

Future Work and Open Ideas

The following directions were raised during the project but not fully explored. They represent natural extensions of the current framework and would form the basis for further research. Note: These were just potential ideas from meetings please read reports and the README files for suggested next steps.

1. Climatology-Augmented Forecasts

Incorporate ERA5 climatological wind distributions as a supplementary forecast source, particularly beyond the 10-day PredictWind forecast horizon. A convex combination of discounted short-range NWP forecasts with climatology-derived arc costs may offer a robust long-horizon routing policy, gracefully degrading as numerical weather prediction skill decays with lead time.

2. Transition-Matrix Climatology

For each ocean grid point, construct a transition matrix from historical ERA5 data capturing the conditional distribution of the next wind state given the current observed wind. This extends climatology from a static prior to a Markovian online update that conditions on what the vessel is currently observing — a potentially powerful complement to the MPC policy that would allow the routing framework to learn from recent wind observations in real time.

3. CVaR Risk-Averse Policies

Implement Conditional Value-at-Risk (CVaR) analysis on both immediate arc costs and cost-to-go estimates. A risk-averse variant of the adaptive policy may be preferable in high-stakes commercial operations where avoiding worst-case voyage delays is more important than minimising expected cost. The trade-off between expected performance and tail-risk protection could be parametrised by a risk-aversion coefficient.

6. Engine Power as a Decision Variable

Formally incorporate engine power as an explicit decision variable rather than a fixed parameter. Discretise or interpolate energy levels and explore the interaction between engine strategy and routing policy. Lower engine power settings create greater differentiation between routing policies and may reveal richer adaptive behaviour, while also being more representative of fuel-optimisation objectives in commercial shipping.

7. Backward DP with Climatological Expectation

For every ocean grid point, compute the expected time-to-go by running backward dynamic programming using climatological wind distributions as the transition model. This produces a climatology-based cost-to-go heuristic that could seed or warm-start the adaptive MPC

policy, providing an informed initial estimate of downstream value before NWP forecasts are available.

8. Full Atlantic Benchmark Under Varying Engine Regimes

Extend the existing benchmark sweep to include multiple engine power levels. Measure how the hindsight gap Δ^p and the policy ranking vary as the wind's contribution to routing decisions increases. Cross-reference findings with prior year project reports to test consistency of results and extend prior hypotheses about the relationship between engine power and routing policy effectiveness.

9. Forecast Verification Expansion

Extend the forecast verification analysis to further lead times and additional meteorological variables (significant wave height, gust speed). Investigate whether forecast skill decay correlates with the periods in which the adaptive policy most strongly outperforms fixed-forecast alternatives. This would provide a mechanistic link between NWP model quality and the value of adaptive routing.

10. Automated GRIB Data Pipeline

Develop a fully automated pipeline to download daily PredictWind GRIB forecasts and stage them for the policy framework. Building on the PyAutoGUI exploration from earlier meetings, this would enable true online rolling-horizon experiments without manual data collection, allowing continuous evaluation of the adaptive policy over extended time periods and larger geographic domains.